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Faculty of Information and Communication Technology (FICT)

UCCN2513 Mini Project

Individual Project – June 2024 trimester

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Tutorial Group: T2

Items	**Marks**
Motivation and problem statement (5%)	
Objective (5%)	
Project scope (4%)	
Contribution of the project (4%)	
Design (3%)	
Implementation (3%)	
Performance Study (3%)	
Conclusion (3%)	
Total (30%)	

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Introduction

1.1 Motivation

In today's fast-paced corporate world, creating a healthy and efficient work environment is essential. Organizations increasingly understand the significant impact that workplace conditions have on employee productivity and well-being, making advanced monitoring solutions a necessity. Traditional office maintenance methods are often manual, inefficient, and prone to errors, leading to inconsistencies in comfort that can affect performance. By integrating Internet of Things (IoT) technology into office monitoring systems, we can enable real-time data collection, analysis, and automation. This project seeks to utilize IoT to create a smart, responsive office environment that enhances both employee comfort and productivity, while also promoting sustainability and energy efficiency. With IoT, we aim to revolutionize office monitoring, ensuring optimal working conditions that adapt to the evolving needs of the modern workforce.

1.2 Problem Statement

Office environments critically influence employee health, comfort, and productivity. However, traditional methods of monitoring office conditions rely heavily on manual data collection, which is time-consuming and prone to errors. These conventional approaches fail to offer the continuous, real-time data necessary to maintain consistent indoor conditions, leading to problems like fluctuating temperatures, varying humidity, and inconsistent air pressure. Such environmental inconsistencies can create discomfort among employees, negatively impacting their well-being and work efficiency. To address these challenges, this project proposes developing an IoT-based environmental monitoring system specifically designed for office settings. By deploying sensors throughout the workplace, the system will gather and analyze real-time data on key environmental factors, enabling proactive adjustments to maintain an ideal indoor environment. This IoT solution will enhance employee health and productivity and contribute to a more sustainable and energy-efficient office environment.

Objective

The goal of this project is to create a comprehensive IoT-based office monitoring system featuring three distinct user interfaces: Reception, Workshop, and Meeting Room. This system aims to enhance office efficiency, comfort, and control. The project will adhere to the SMART criteria, focusing on specific, measurable, achievable, relevant, and time-bound objectives.

Specific

Design and implement a system to automate and control key office functions, including door operation, CCTV activation, environmental monitoring (temperature, humidity, pressure, smoke level), and equipment management (air conditioning, lighting, Wi-Fi, computers) across three user interface sections: Reception, Workshop, and Meeting Room.

Measurable

The success of the system will be evaluated based on its accuracy in detecting and responding to environmental changes, managing equipment through sensor data, and providing real-time notifications and controls via MQTT, Node-RED, and Telegram. Key performance metrics will include response time, accuracy of sensor readings, and user satisfaction with the system's usability.

Achievable

Utilize established IoT frameworks and tools, such as Node-RED, InfluxDB, MQTT, and Telegram, to develop a fully integrated and functional system. The project will simulate real office conditions to ensure effective management of environmental parameters and equipment controls.

Relevant

The project addresses the increasing demand for smart office solutions that enhance energy efficiency, employee comfort, and workplace automation. It aligns with current trends in IoT, smart building technologies, and sustainable office practices.

Time-bound

The project will be completed within a 10-week timeframe, beginning on June 19, 2024, and concluding on August 21, 2024. This schedule includes milestones for design, development, testing, and deployment to ensure timely progress and successful achievement of the project's objectives.

Project Scope

This project involves creating a scalable and modular IoT-based office monitoring system designed to simulate and manage various environmental parameters across multiple rooms or zones within an office setting. The system will integrate virtual sensors, an IoT platform, a database, and user interfaces accessible via handheld devices such as tablets and mobile phones, as well as through web access.

The project encompasses the collection, analysis, and visualization of environmental data—such as temperature, humidity, smoke levels, sound levels, and pressure—as well as the implementation of control mechanisms to manage these conditions. Users will interact with the simulation through intuitive interfaces, enabling them to control environmental settings and observe system responses in real-time.

The scope is confined to the development of simulation software, without involving physical hardware. It will utilize Node-RED for generating simulated sensor data, InfluxDB for historical data storage, and Telegram for notifications and remote control capabilities. The system will be designed to work within a simulated environment using virtual sensors and emulators rather than physical devices.

In particular, the project will employ Node-RED to generate and manage random sensor data, which will be processed and visualized on a Node-RED dashboard. Historical data will be stored in InfluxDB for ongoing analysis. The system will feature a comprehensive dashboard with interactive controls, allowing users to manage lighting, air conditioning, and other office equipment based on simulated environmental conditions.

A key component will be the integration of MQTT for real-time data transmission and control commands. Additionally, a Telegram bot will be developed to provide users with remote access to environmental monitoring and control functionalities.

The project will focus on ensuring the system's scalability and adaptability for future upgrades or integration with additional office systems. The simulation will be thoroughly tested to evaluate its performance, accuracy, and effectiveness in replicating a smart environmental monitoring setup. This approach will enable the creation of a robust and flexible office monitoring system entirely through software simulation.

Contribution of the Project

This project makes a substantial contribution to the field of office environmental monitoring by pioneering a sophisticated and adaptable IoT-based simulation system. The system innovatively employs virtual sensors and emulators to accurately replicate and manage environmental parameters—such as temperature, humidity, smoke levels, and pressure—across various rooms or zones within an office setting. This approach eliminates the need for physical hardware, making the system both cost-effective and versatile.

By leveraging advanced technologies like Node-RED for dynamic data generation, InfluxDB for comprehensive historical data storage, and MQTT for seamless real-time data transmission, the project enhances both the efficiency of data handling and the robustness of control mechanisms. The integration of these technologies ensures that the system operates with high precision and reliability.

The user interfaces—accessible via tablets, mobile phones, and web browsers—offer an intuitive and user-friendly platform for real-time interaction, control, and data visualization. This ease of access empowers users to actively monitor and manage office conditions with minimal effort, improving both operational efficiency and user experience. The inclusion of Telegram for remote notifications and control further extends the system's functionality, allowing users to receive alerts and make adjustments from virtually anywhere.

The project's design emphasizes scalability and adaptability, ensuring that the system can evolve to accommodate future upgrades and integrate seamlessly with additional office systems or technologies. This forward-thinking approach positions the system as a valuable asset for modern office environments.

Furthermore, the thorough testing phase will ensure that the system meets high standards of performance, accuracy, and effectiveness, validating its reliability as a comprehensive environmental monitoring solution. Beyond its immediate applications, the project provides a robust framework for future research and development in IoT and environmental monitoring, offering valuable insights and methodologies that can inspire and inform subsequent innovations in the field. The system's flexible and scalable design not only addresses current needs but also anticipates future demands, making it a significant advancement in the realm of smart office technology.

Design

5.1 System Architecture

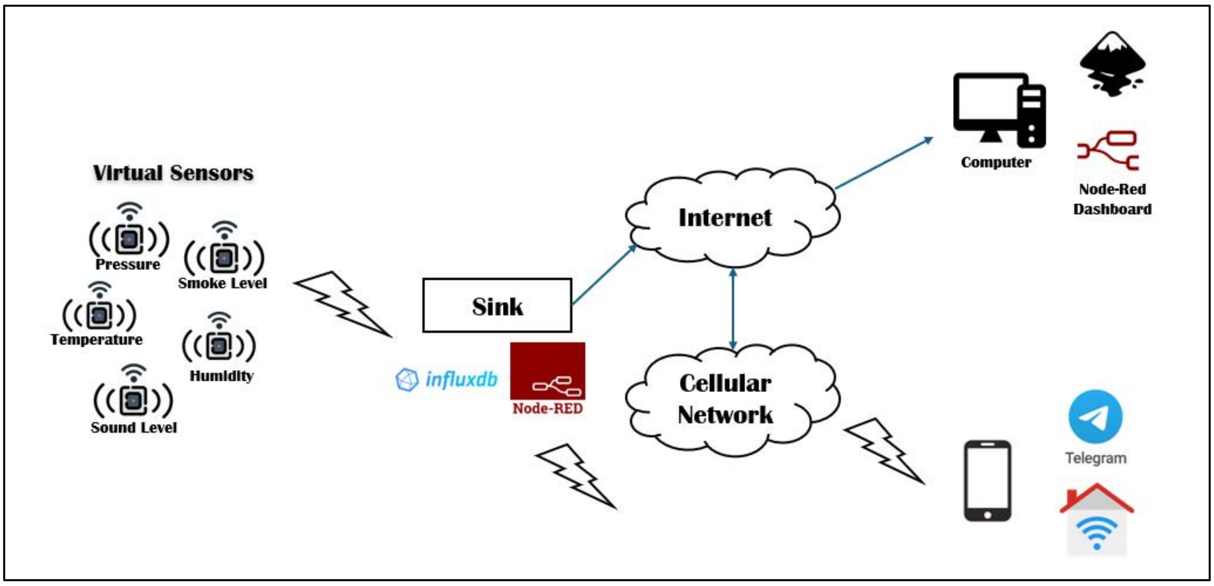


Figure 5.1 – System Architecture Diagram

The system architecture diagram provides a high-level overview of the system for managing virtual sensors in an office monitoring project. Node-RED generates virtual sensor data through a function node, which simulates real sensor outputs. This data is then processed and controlled within Node-RED, which also integrates with an SVG UI designed in Inkscape for visual representation. Processed data from Node-RED is stored in InfluxDB for historical analysis and retrieval. Additionally, Node-RED facilitates communication with the IoT MQTT Panel and Telegram for real-time data exchange and remote control. The diagram demonstrates the central role of Node-RED in generating, processing, visualizing, and communicating sensor data within the system.

5.2 Project Flow

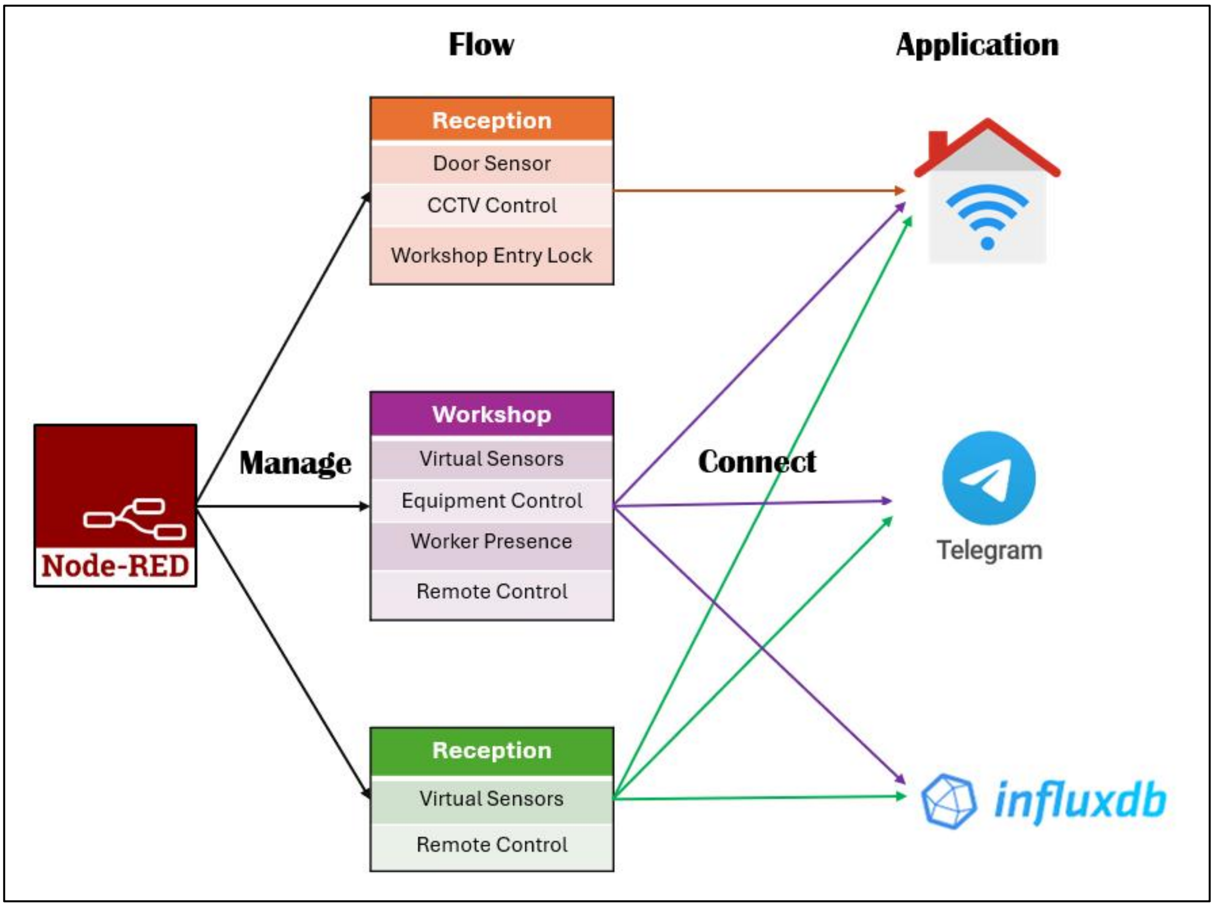
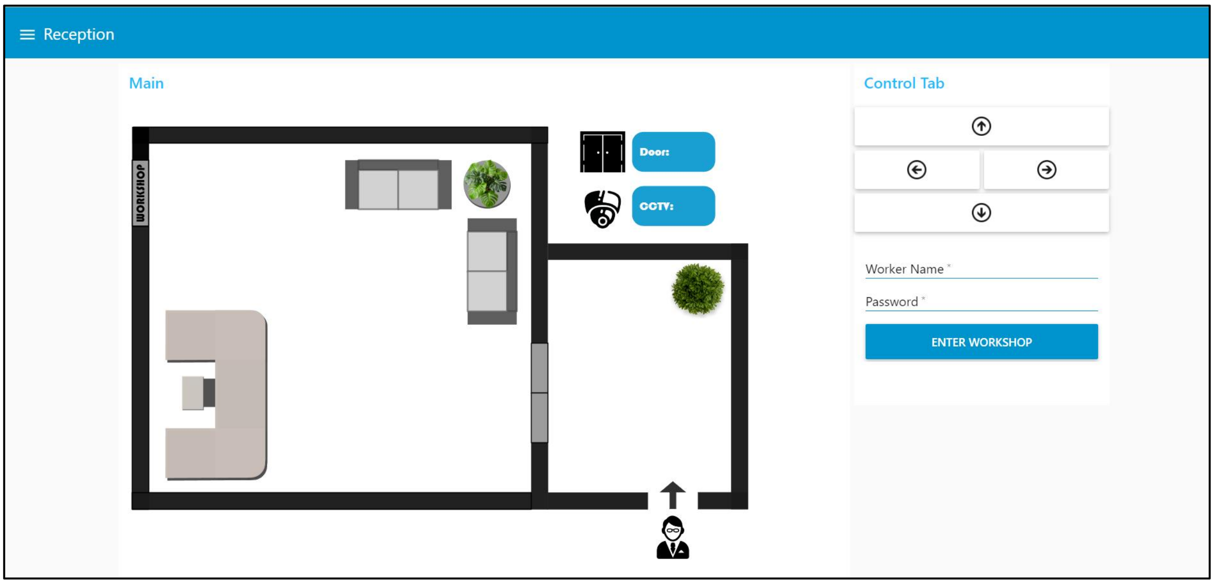


Figure 5.2 – Project Flow Diagram

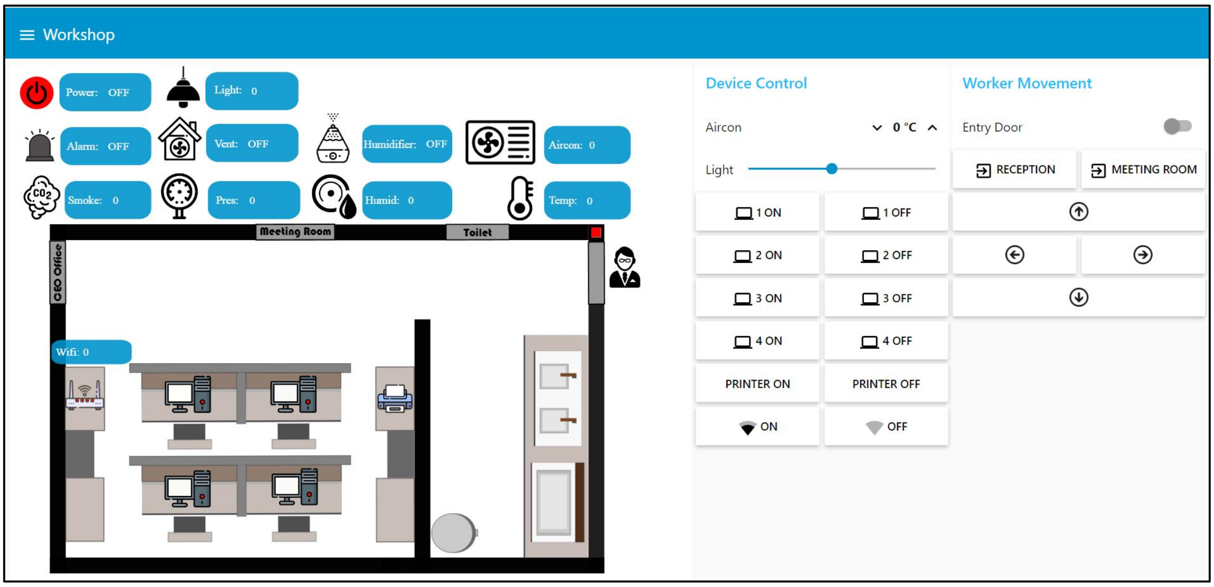
The project flow diagram for the IoT office monitoring system is divided into three sections: Reception, Workshop, and Meeting Room. In Reception, door sensors and CCTV manage entry and monitoring, with updates sent to the MQTT dashboard. In the Workshop, Node-RED generates and uses simulated environmental data to control equipment like air conditioning and alarms, with remote management via MQTT and Telegram. Data is stored in InfluxDB. In the Meeting Room, lighting, air conditioning, and sensors are automated based on presence, with additional remote controls for video calls, also using InfluxDB for data storage. The diagram shows the integration of these components through MQTT and Telegram for communication and InfluxDB for data storage.

5.3 Node-RED UI Interfaces

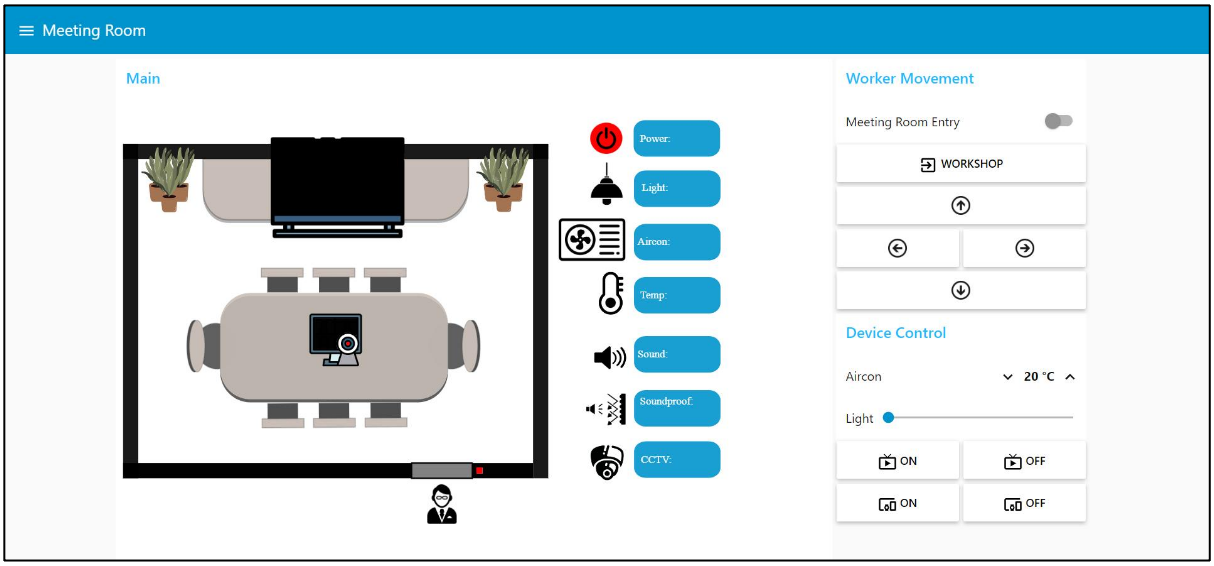
<u>Reception</u>



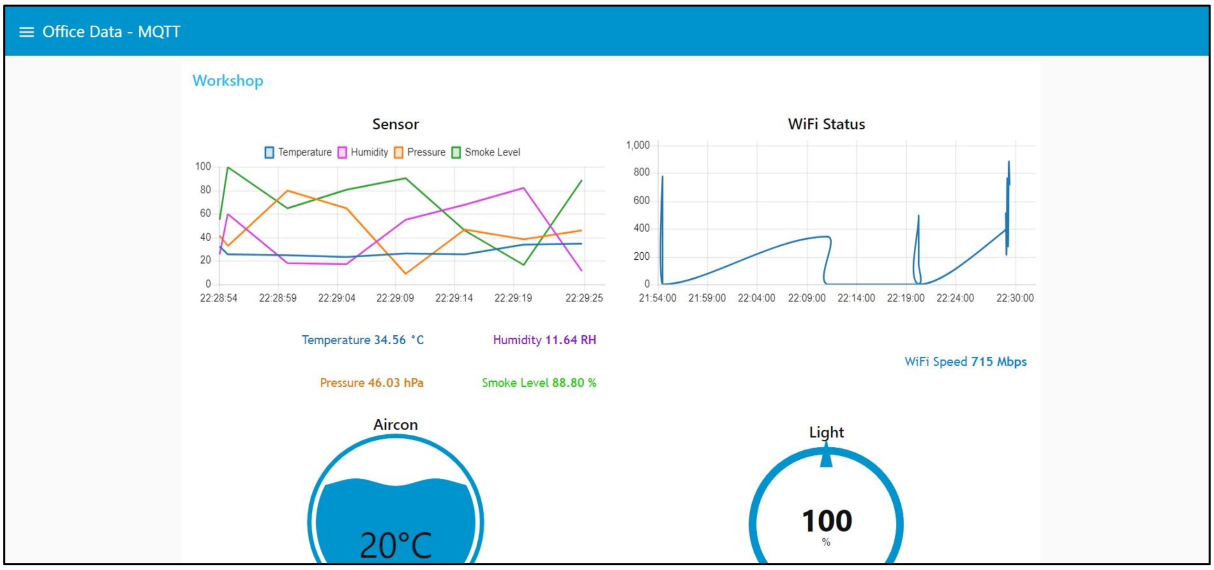
<u>Workshop</u>

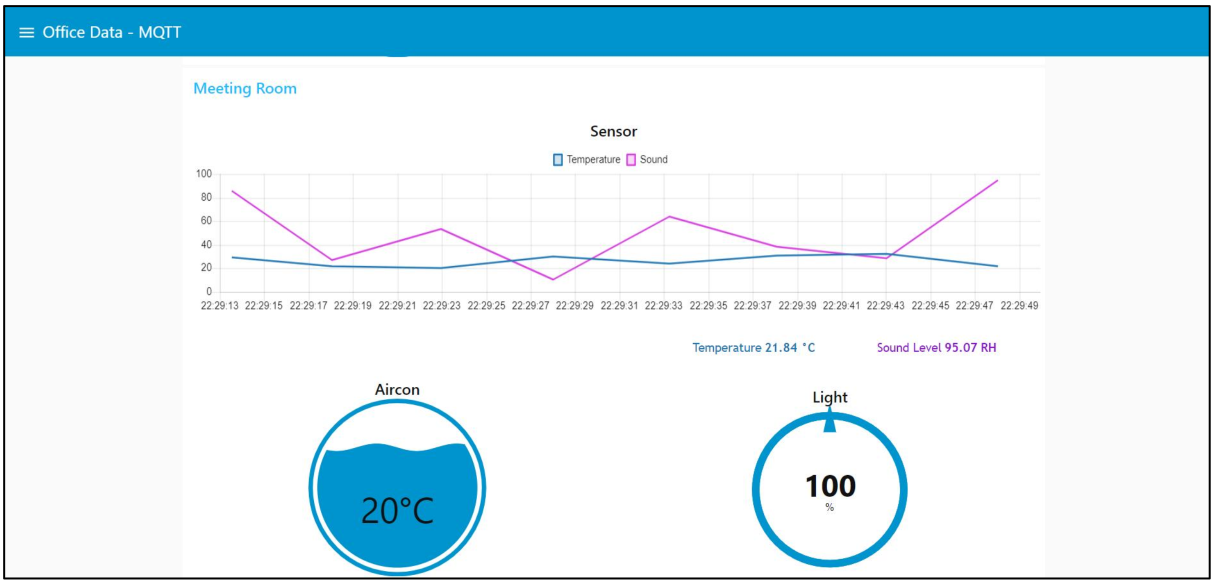


<u>Meeting Room</u>

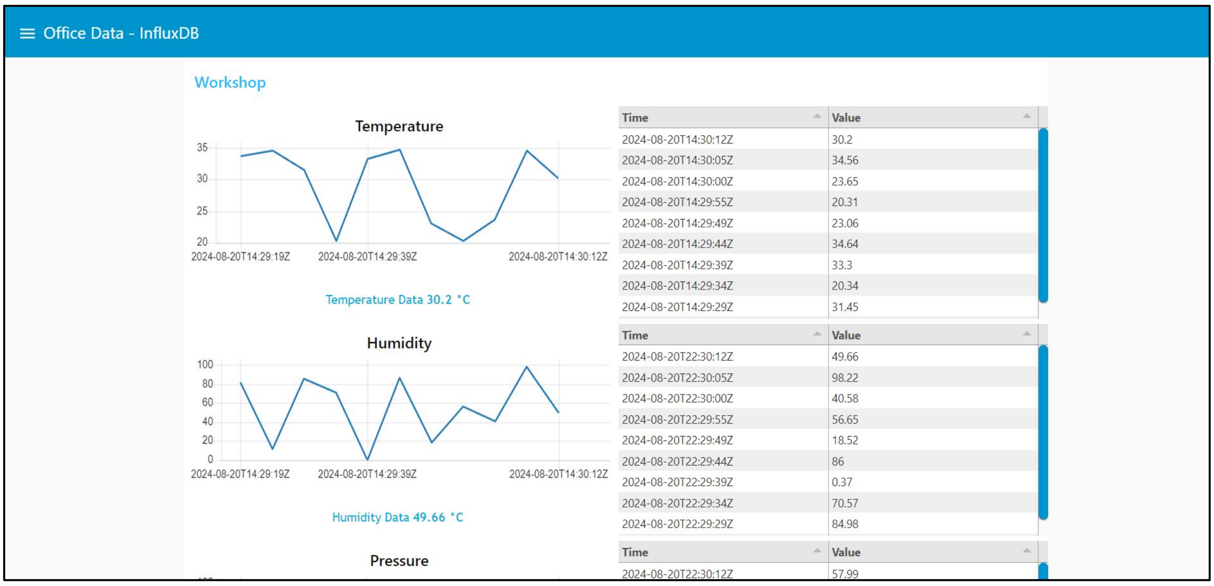


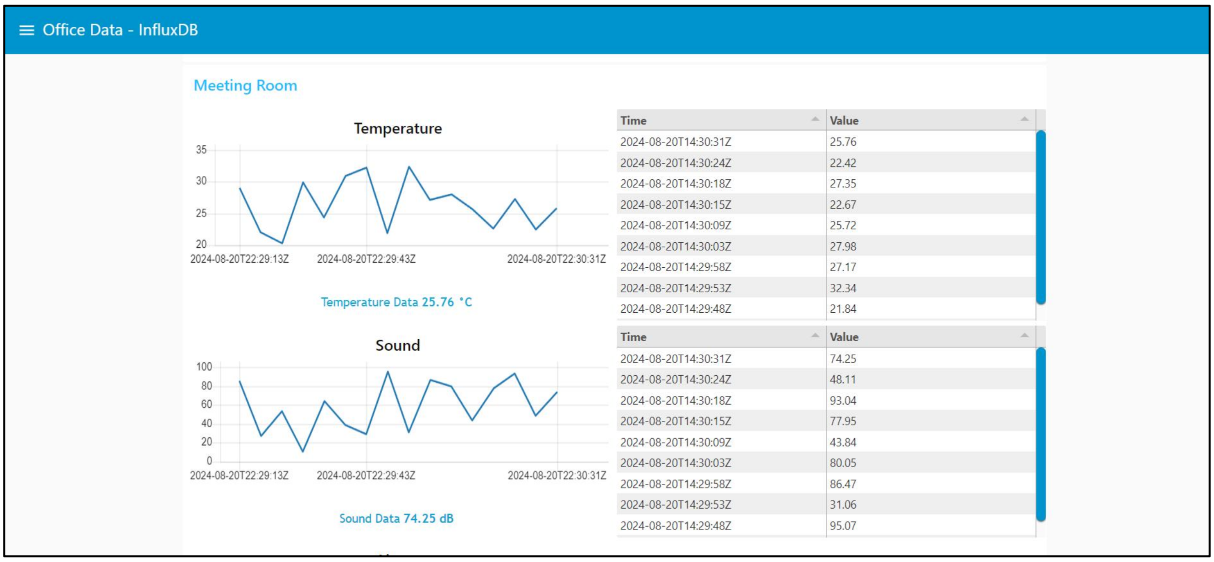
Office Data – MQTT





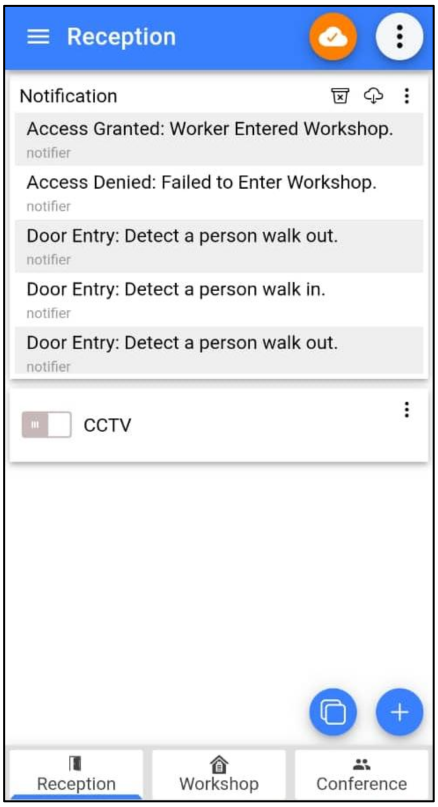
Office Data – InfluxDB



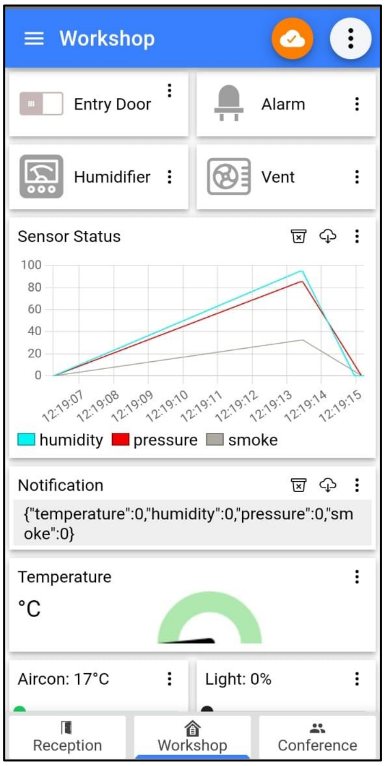


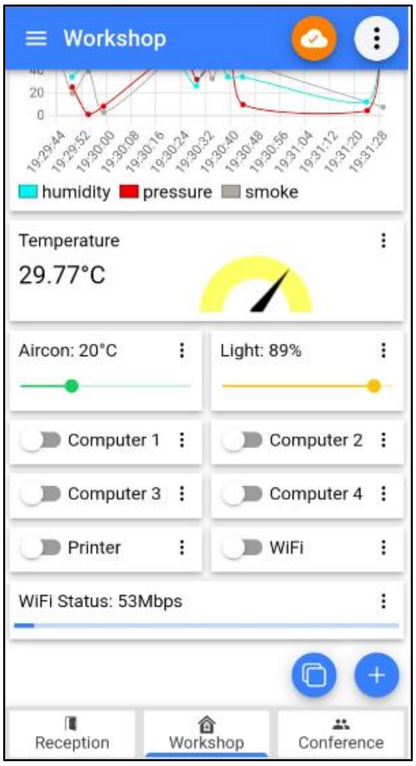
5.4 MQTT Dashboards

Reception

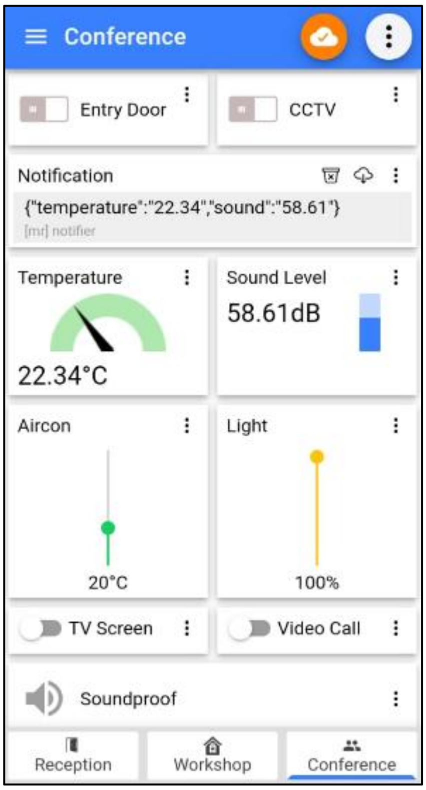


Workshop





Meeting Room



Implementation

6.1 Startup and Installation

To initiate IoT office monitoring system, start by installing the necessary software components. First, install **Node-RED** on server or local machine which will process sensor data and manage system workflows. Next, install **InfluxDB** to handle data storage, set up the required databases, and retention policies within InfluxDB to efficiently manage and store sensor data. Install an MQTT broker, such as **IoT MQTT Panel**, to facilitate communication between Node-RED, your dashboards, and other system elements. Configure a **Telegram bot** to enable remote control and notifications, integrating it with Node-RED for seamless interaction. Use **Inkscape** to design and create user interfaces, ensuring they are visually appealing and functional for your monitoring system.

6.1.1 Necessary Nodes Setup in Node-RED

Install necessary nodes in Node-RED Manage Palette, such as *node-red-dashboard* for UI components, *node-red-contrib-mqtt* for MQTT communication, *node-red-contrib-influxdb* for InfluxDB integration, and *node-red-contrib-telegrambot* for Telegram interaction.

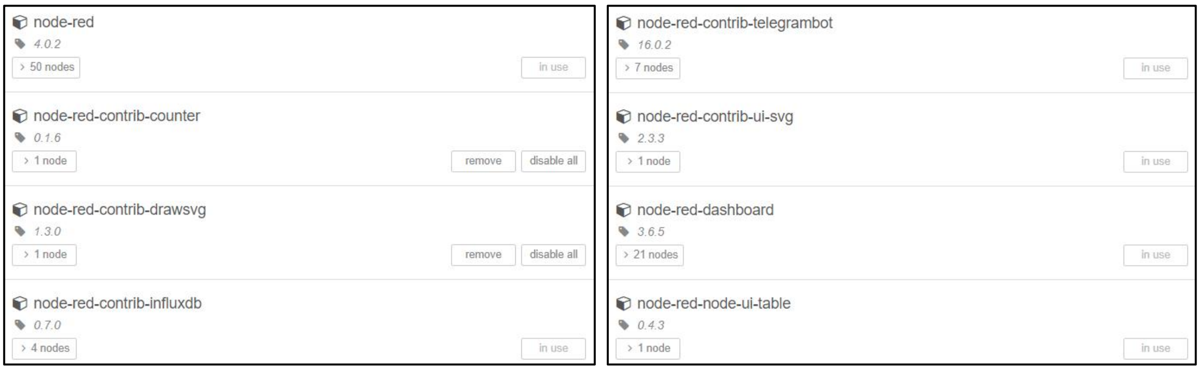


Figure 6.1.1 – Node-RED Palettes

6.1.2 InfluxDB Setup

Set up InfluxDB by creating a bucket for historical data storage and obtaining an API token for Node-RED to securely write data to the database.

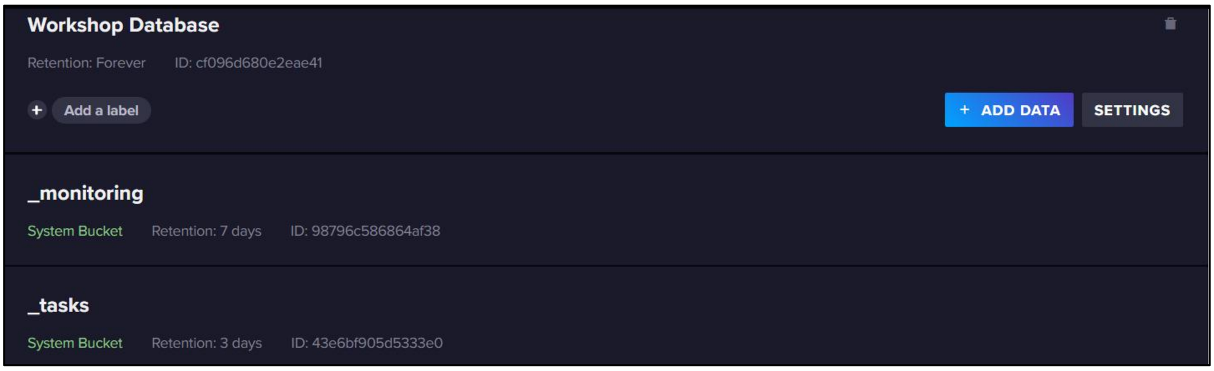


Figure 6.1.2 – InfluxDB Bucket Creation

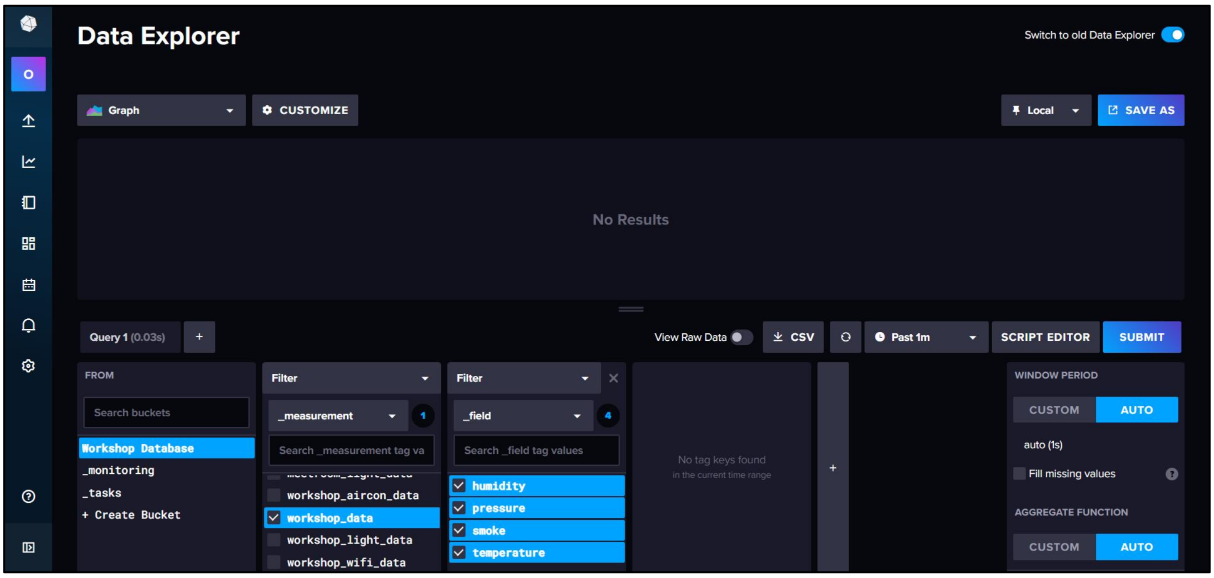


Figure 6.1.3 – InfluxDB Data Explorer

6.1.3 Software Setup for Telegram

To set up a Telegram chatbot, first create a Telegram account if you don't already have one. Search for *BotFather* in Telegram and start a chat by typing */start*. Next, enter */newbot* to begin creating your chatbot. Once the chatbot is created, BotFather will provide a token. Save this token, as you'll need it for future use. Lastly, search your own bot by typing *@username_bot*.

Figure 6.1.4 – Telegram BotFather

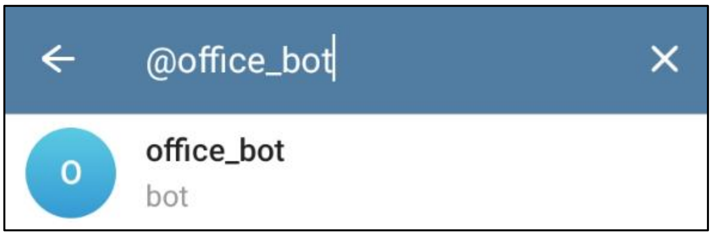


Figure 6.1.5 – Telegram Own Bot

6.1.4 Software Setup for IoT MQTT Panel Mobile Access

To set up the IoT MQTT Panel, start by adding your MQTT broker connection details, including the broker address and port. Next, design your dashboards by adding and configuring panels to display and control your IoT devices. Link these panels to relevant MQTT topics for real-time data exchange and control.

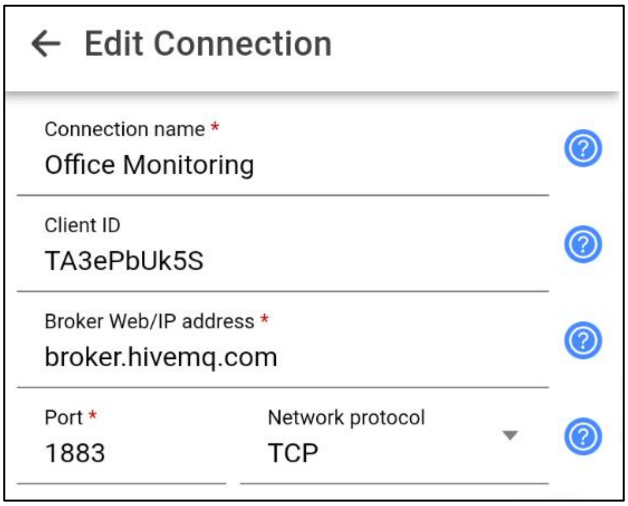


Figure 6.1.6 – Add MQTT Connection

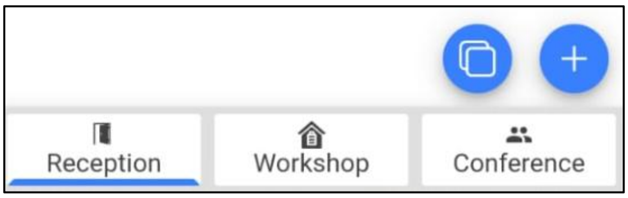
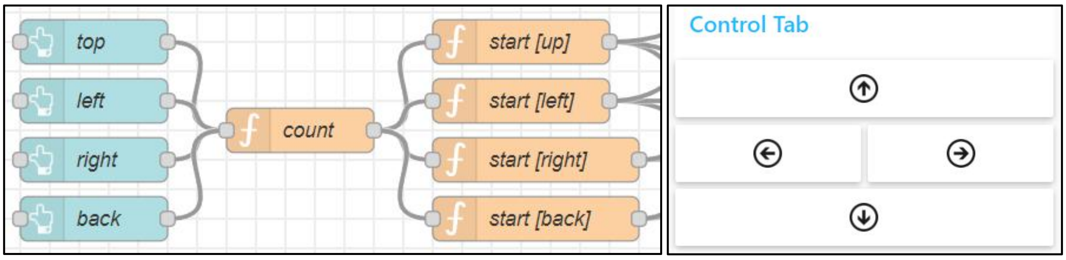


Figure 6.1.7 – Add MQTT Dashboards and Panels

6.2 Configuration

6.2.1 Node-RED Flow Configuration

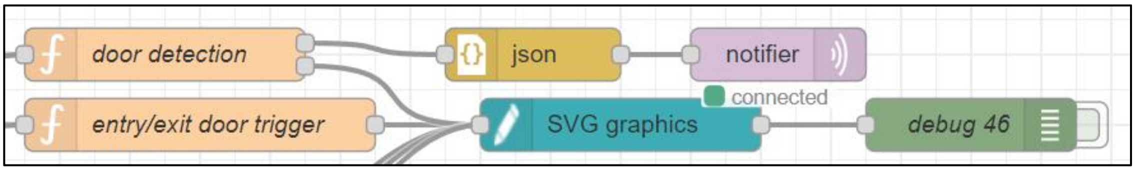
First, design flows for Reception, Workshop, and Meeting Room. Use a control tab with directional buttons to simulate worker movement, updating coordinates in the flow context. This triggers automated actions such as opening doors, activating sensors, and adjusting equipment based on the worker's location. Connect nodes to simulate sensor data, control equipment, and manage worker presence.



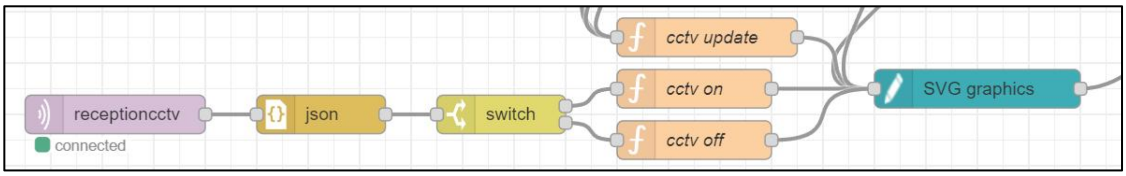
Worker Movement Configuration

6.2.2 Reception Configuration

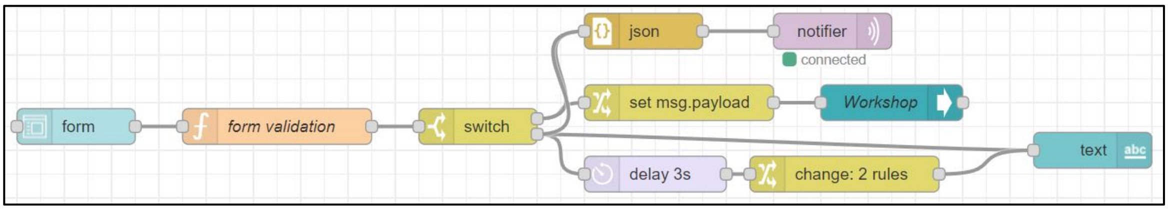
In Reception, the flow manages door sensors, CCTV, and access validation based on simulated worker movement. The door opens and closes automatically, with status updates sent to the MQTT dashboard. CCTV activates automatically upon worker entry, with manual control available via MQTT. Access is controlled through a UI form, granting or denying entry and notifying the dashboard accordingly. If access fails, an alert is triggered, and the dashboard switches to the Workshop view.



Door Sensor Configuration



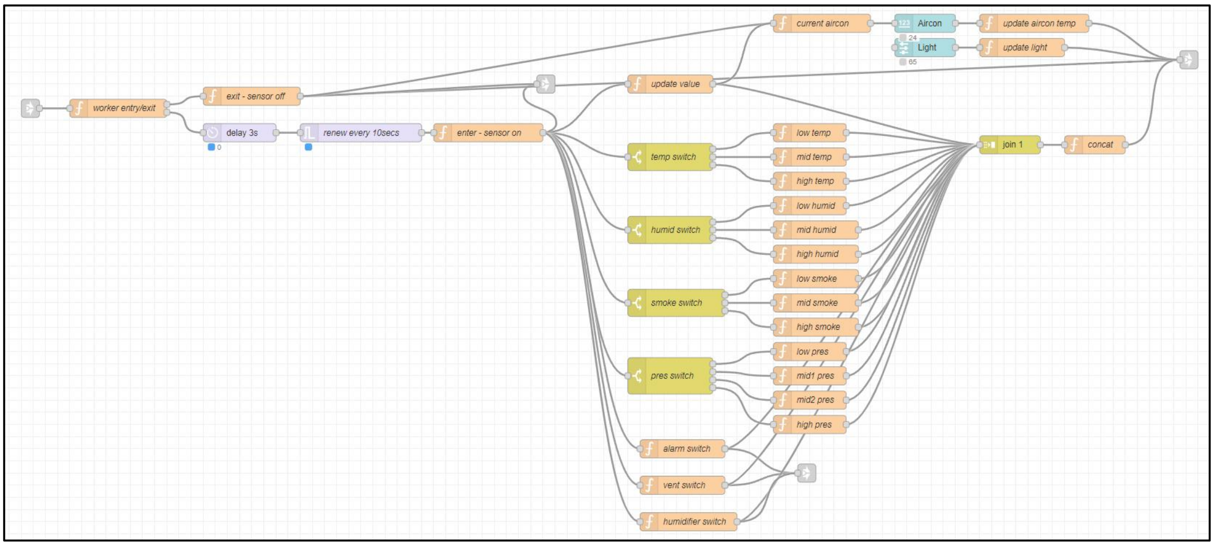
CCTV Control Configuration



Access Validation Configuration

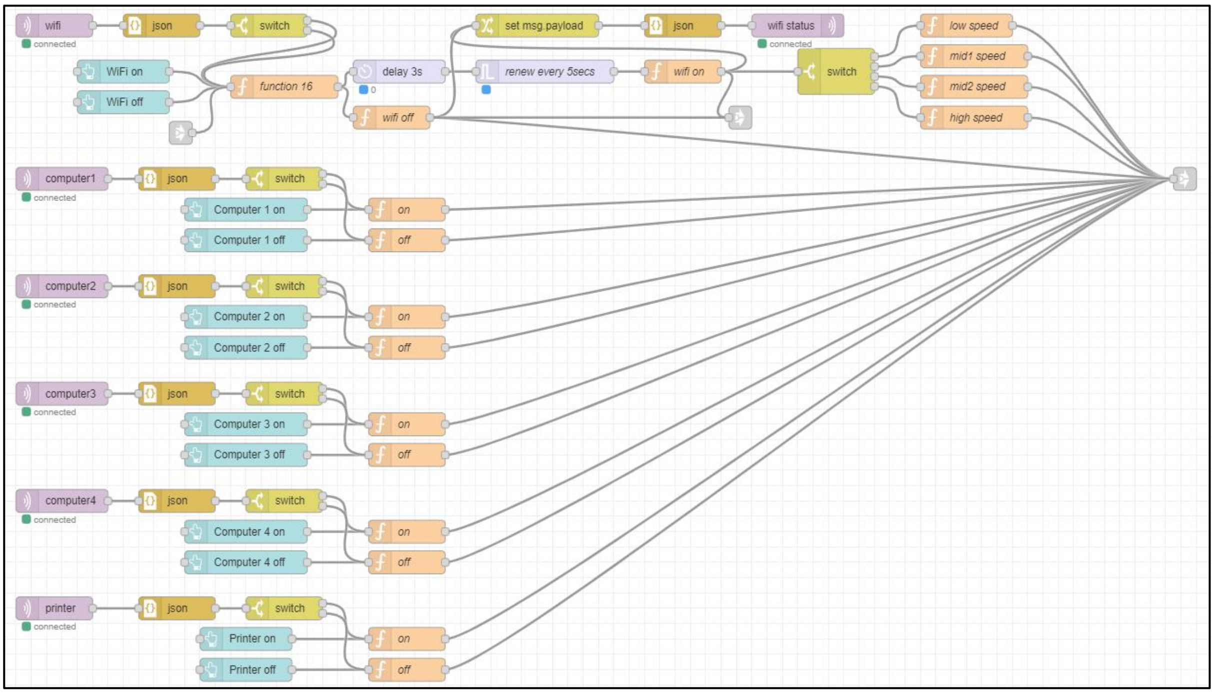
6.2.3 Workshop Configuration

In Workshop, the system simulates sensor data for temperature, humidity, pressure, and smoke levels, updating every 10 seconds. These data, stored in InfluxDB, drives automated equipment control: adjusting air conditioning, activating a humidifier, opening vents, and triggering alarms. Workers can manually adjust air conditioning and lighting via a control tab or MQTT panel. Equipment automatically activates when a worker enters and deactivates when they leave, with changes published to MQTT for remote monitoring and control.



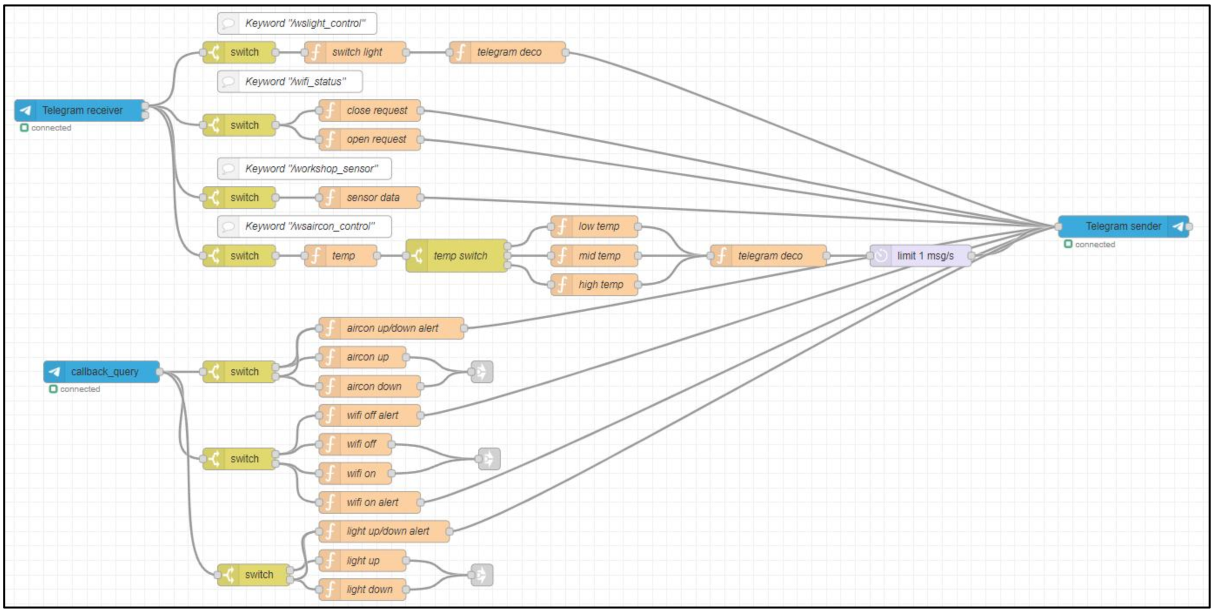
Worker Presence – Automatic Sensors and Equipment Control Configuration

Workers can manually control the air conditioning, lighting, computer, printer, and Wi-Fi via a control tab or the MQTT panel when present in the Workshop. When Wi-Fi is opened, its speed is displayed on the MQTT panel or Node-RED dashboard.

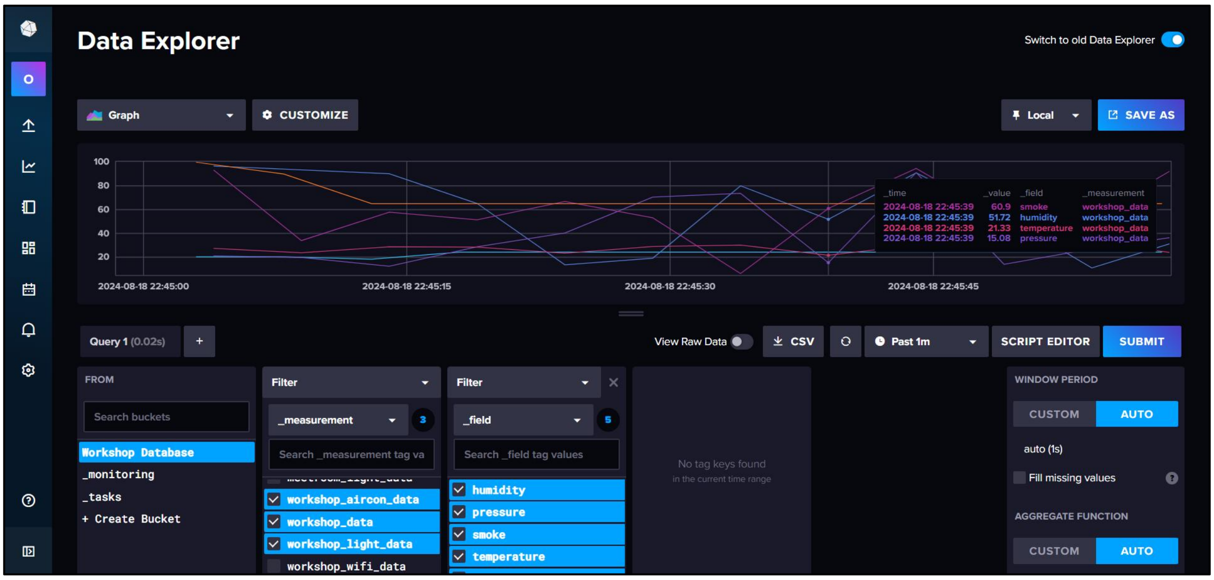


Device Manual Control Configuration

Additionally, a Telegram bot provides workers with the ability to retrieve sensor values and remotely control the lighting and air conditioning. The bot also allows workers to issue commands to open or close the Wi-Fi.



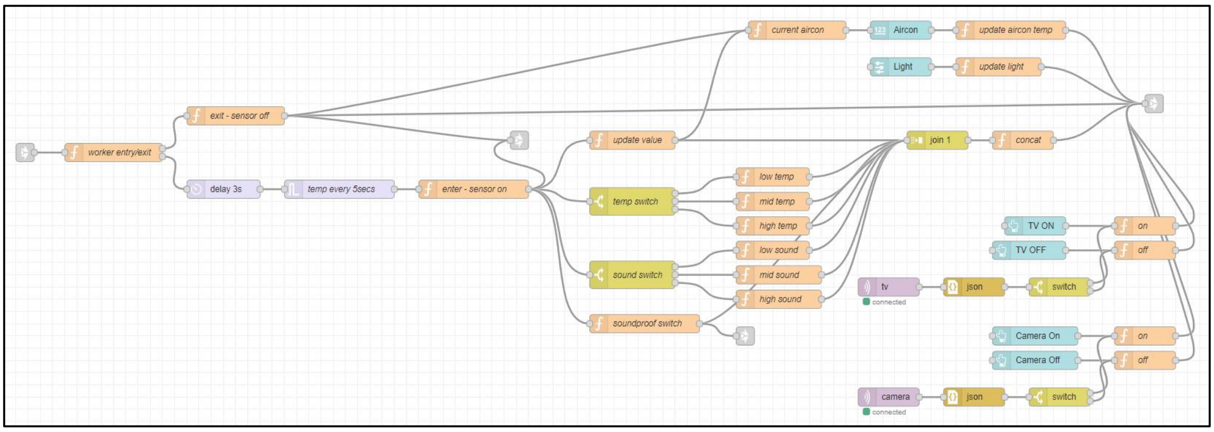
Telegram Configuration



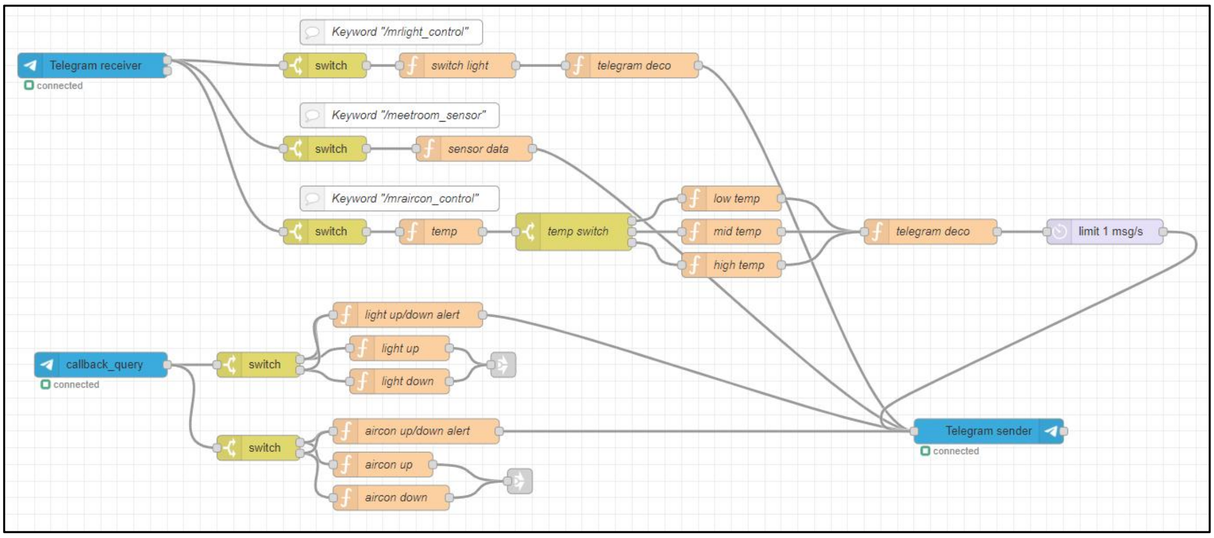
InfluxDB – Workshop

6.2.4 Meeting Room Configuration

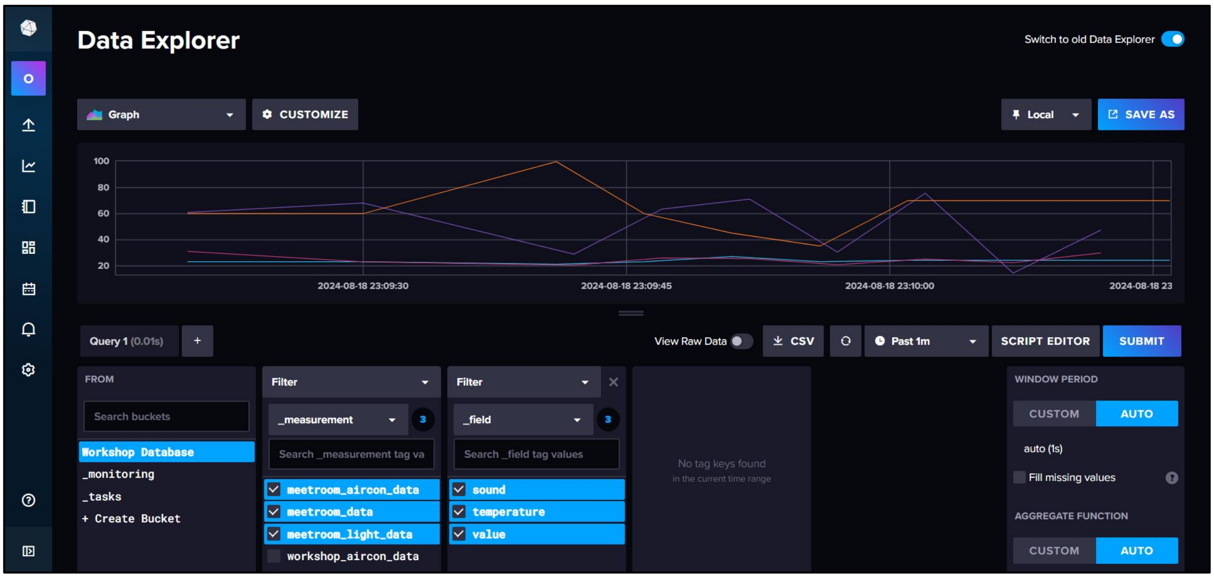
In Meeting Room, temperature and sound sensors activate upon worker entry. Air conditioning, lighting, and TV/video call controls are adjustable via the control tab, MQTT panel, or Telegram bot. All data is stored in InfluxDB, with automatic equipment control based on presence to conserve resources. Changes are published to MQTT for real-time monitoring and remote control.



Worker Presence – Automatic Sensor and Equipment Control Configuration



Telegram Configuration

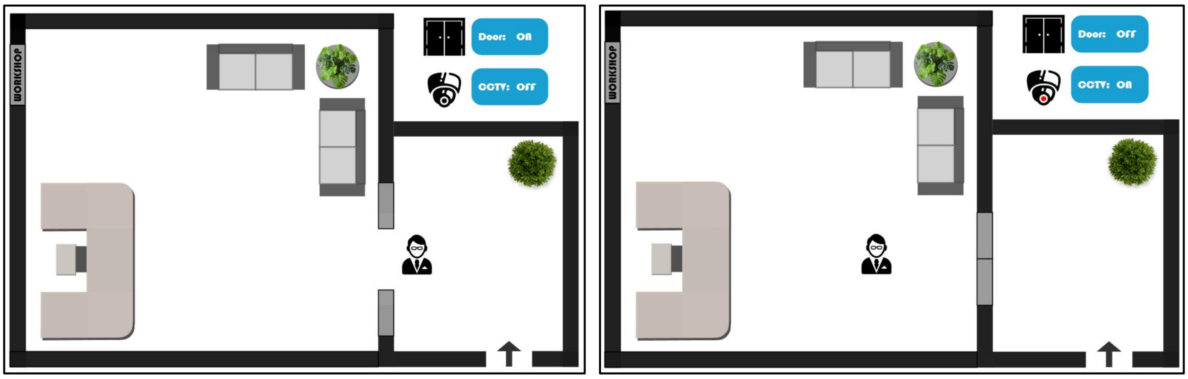


InfluxDB – Meeting Room

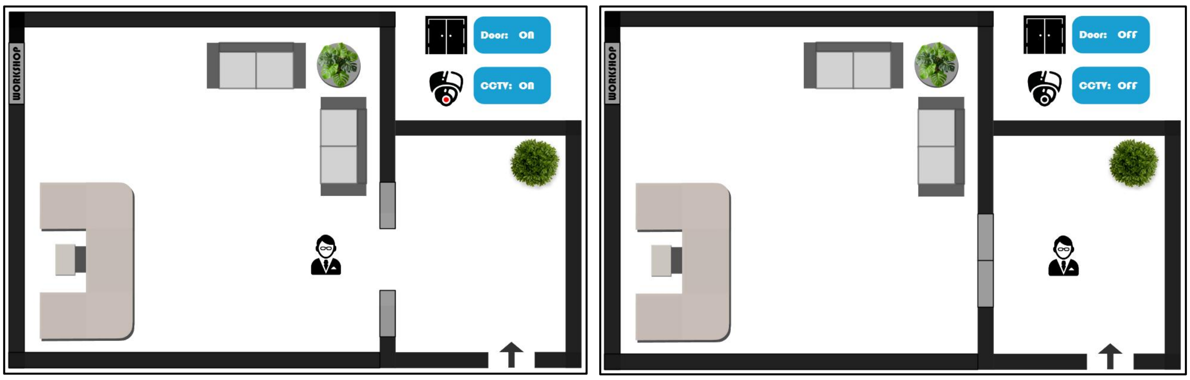
6.3 Operation

6.3.1 Reception Dashboard

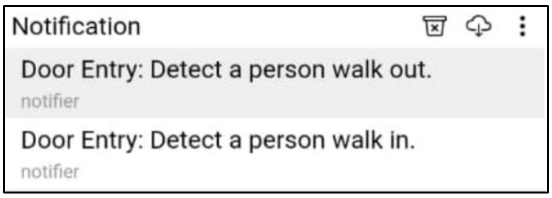
First, the worker uses directional buttons on the control tab to simulate movement. As they approach, the door opens automatically and closes upon entry. The system notifies the MQTT broker and activates CCTV. Second, the worker must complete an access form to worker entry; correct submission grants access to the Workshop dashboard with an MQTT confirmation, while incorrect submission keeps them in Reception and triggers an MQTT notification about denied access.



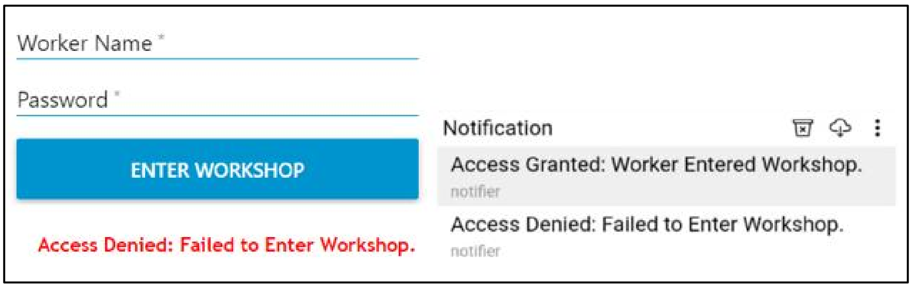
Door Sensor – Enter



Door Sensor – Exit



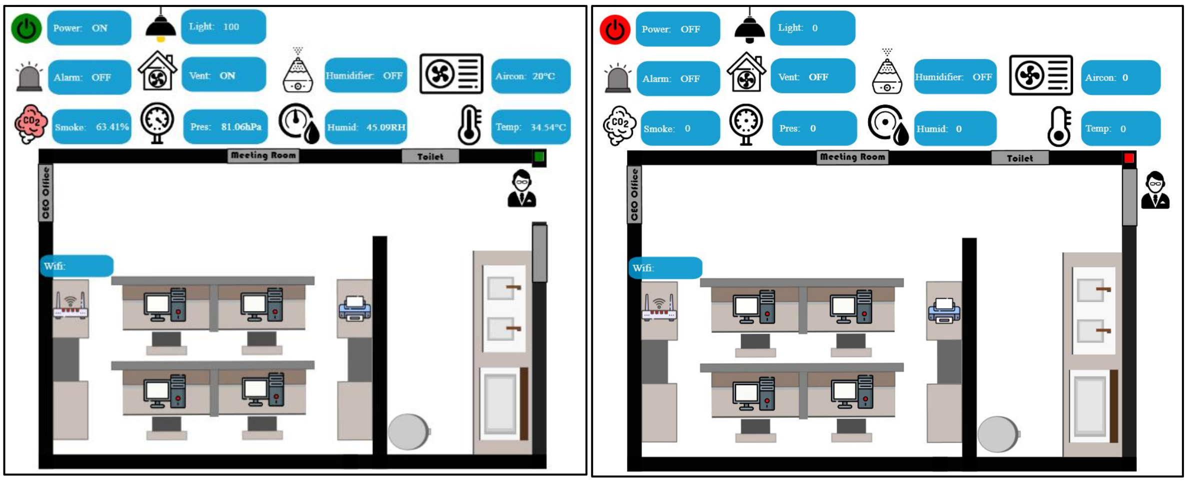
MQTT Notification – Door Sensor



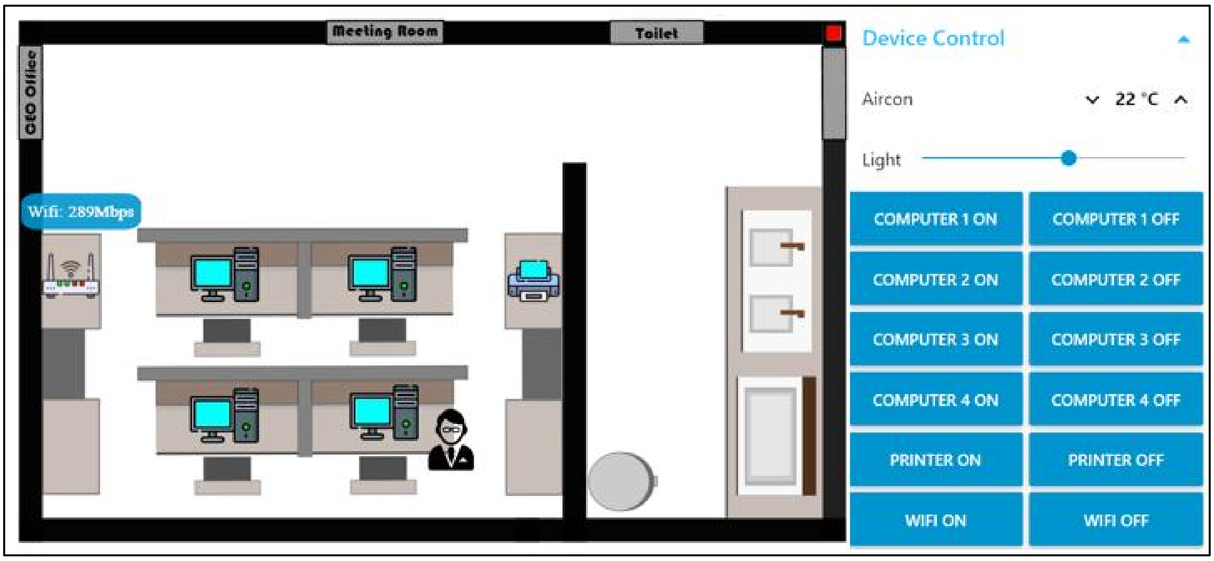
MQTT Notification – Form Validation

6.3.2 Workshop Dashboard

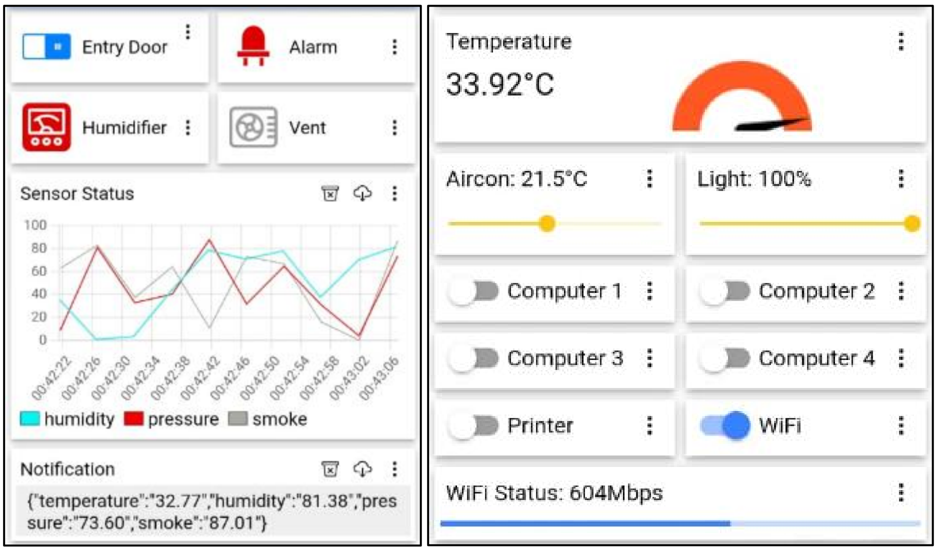
Upon entering the Workshop, the worker gains control over door access through the control tab or MQTT panel. The system automatically activates sensors (temperature, humidity, pressure, and smoke levels) and adjusts equipment like the humidifier, vent, and alarm based on readings. Lights and air conditioning are turned on by default, with options for manual adjustment. Workers can also control Wi-Fi, the computer, and the printer via the control tab or MQTT panel. Sensor data, air conditioning settings, light levels, and Wi-Fi speed are sent to MQTT and stored in InfluxDB. To switch to the Meeting Room or return to the Reception area, the worker can use the control tab. The Telegram bot allows the worker to view sensor data and control the air conditioning with specific commands.



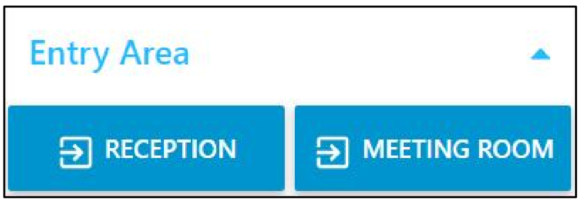
Automatic Sensors



Device Control

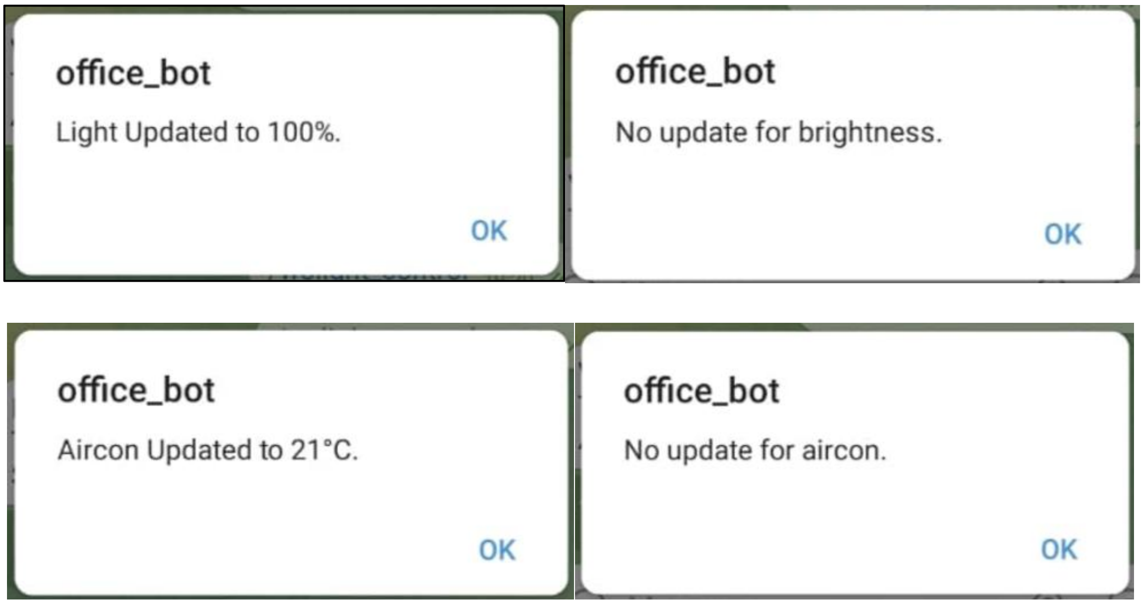


MQTT Dashboard – Workshop



Entry Area

Telegram Bot – Workshop

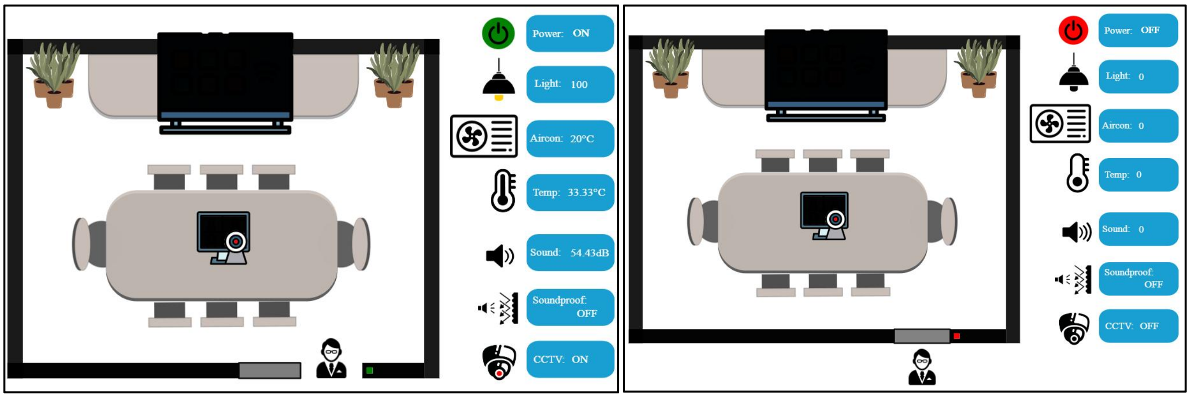


Telegram Bot – Updated Message

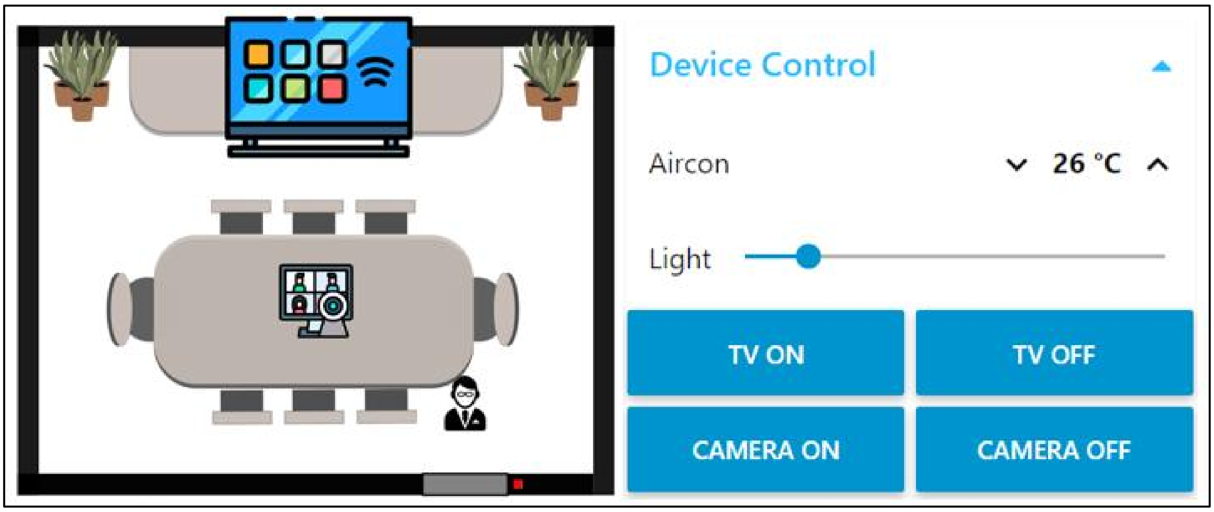
(Based on Worker Presence)

6.3.3 Meeting Room Dashboard

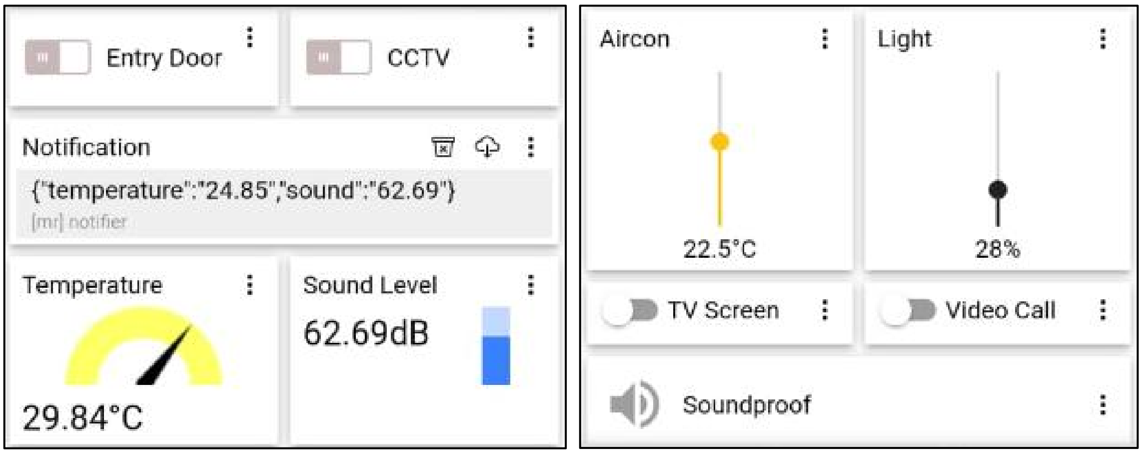
In the Meeting Room, the worker can manually control the entry door via the control tab or MQTT panel. Upon entry, sensors for temperature and sound level are activated, with the system adjusting air conditioning and soundproofing based on sound levels. Default settings for lights and air conditioning are applied, and CCTV is activated, with manual control also available. Data including sensor readings, air conditioning settings, light levels, and CCTV status are sent to MQTT and stored in InfluxDB. Workers can adjust air conditioning, lights, TV screen, and video calls using the control tab or MQTT panel. The Telegram bot enables viewing sensor data with "*/meetroom_sensor*" and adjusting light brightness with "*/light_control*".



Automatic Sensors



Device Control



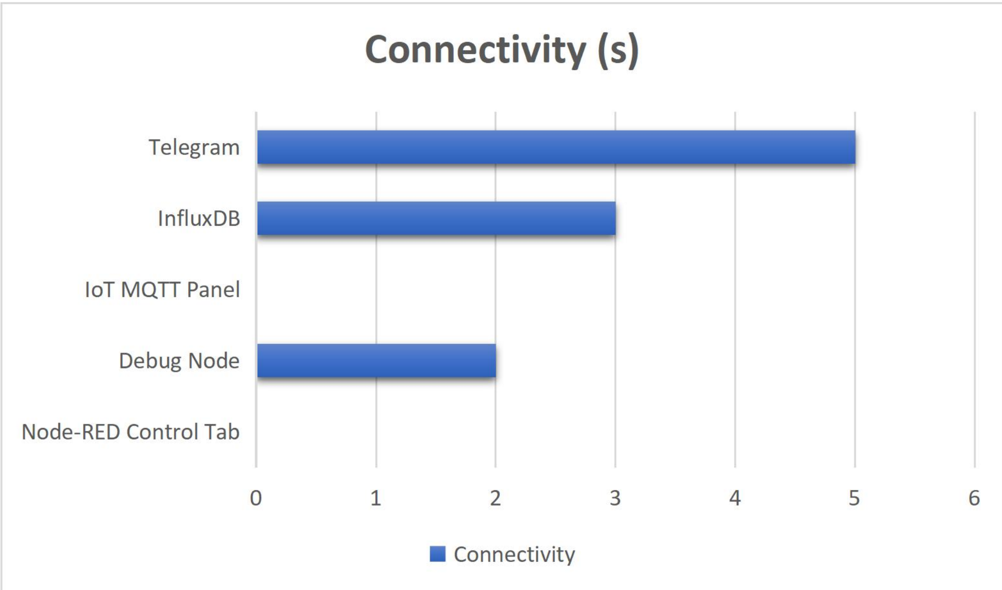
MQTT Dashboard

Telegram

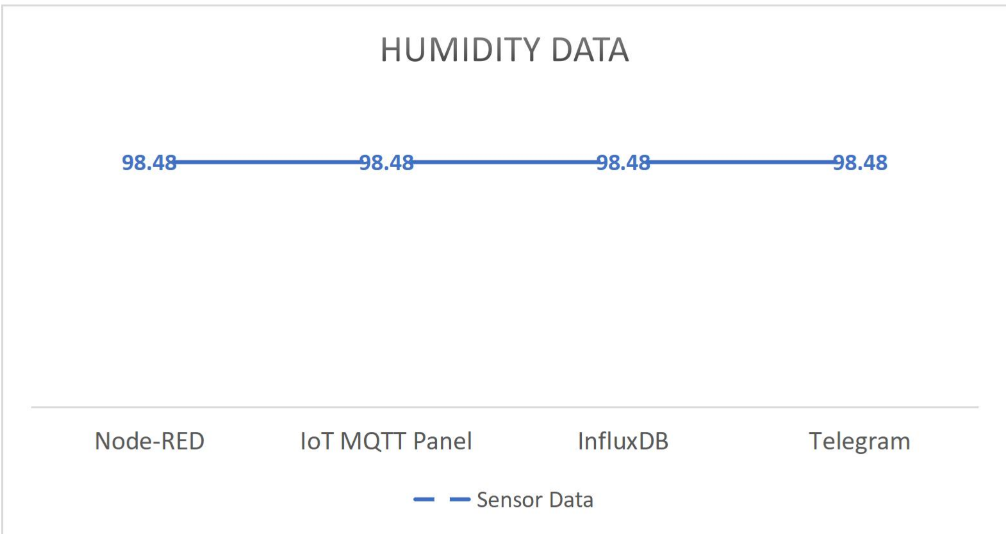
Performance Study

7.1 Data Latency

Device	Node-RED Control Tab	Debug Node	IoT MQTT Panel	InfluxDB	Telegram
Time	20:47:35	20:47:37	20:47:35	20:47:38	20:47:40

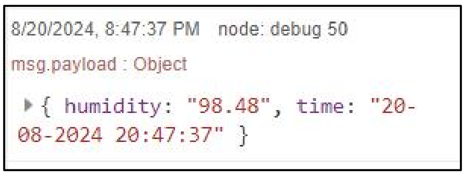


7.2 Consistent Data for All Platforms

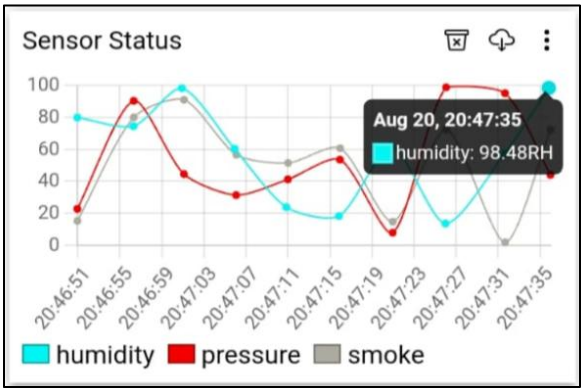


Example of Retrieving Humidity Data

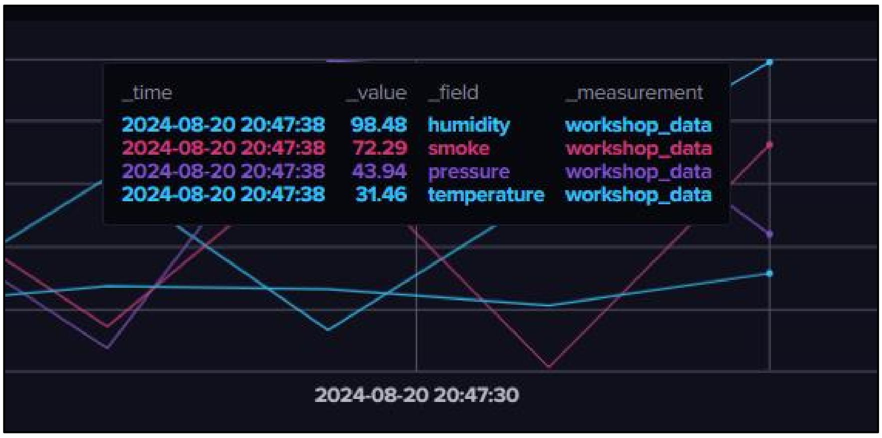
Debug Node



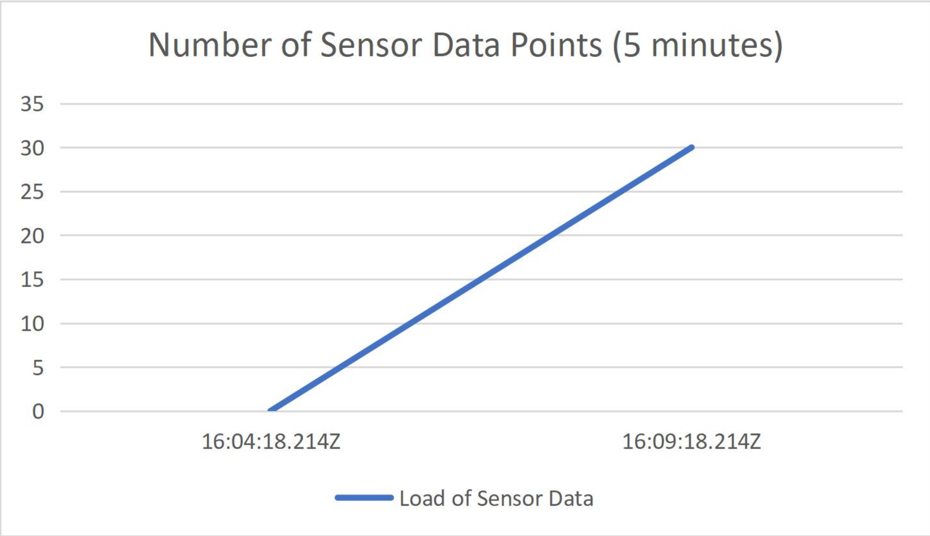
IoT MQTT Panel



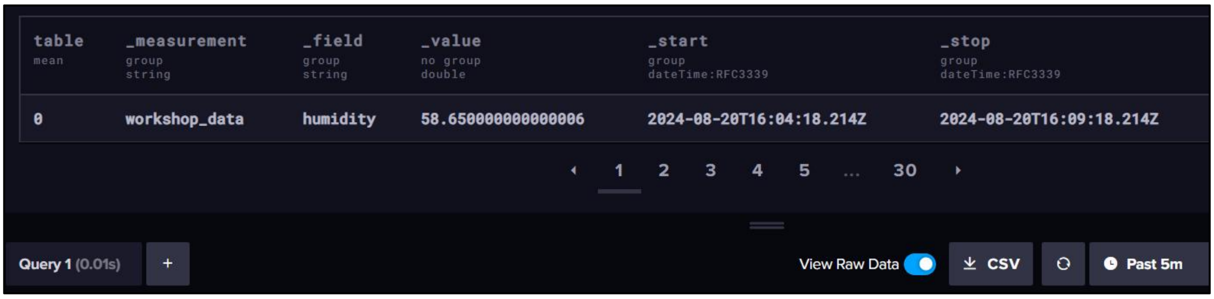
InfluxDB



7.3 Data Scalability



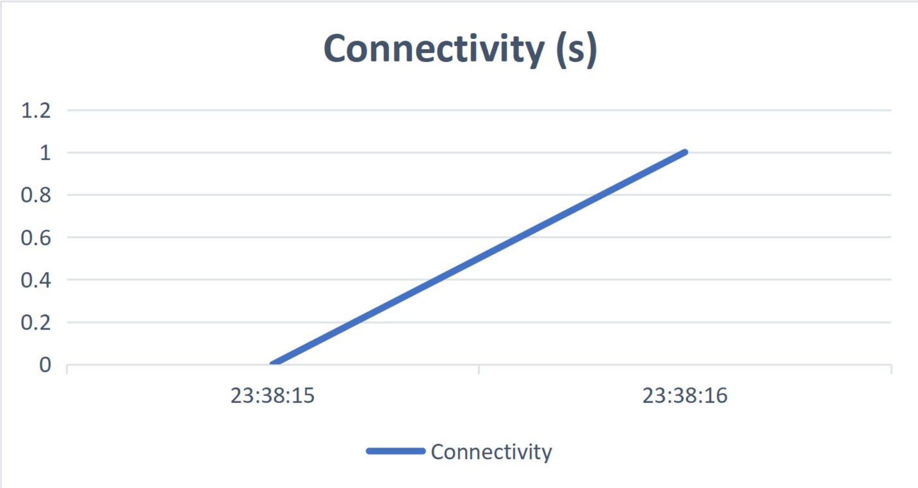
<u>30 Sensor Data Points in every 5 minutes</u>



7.4 Platform Connectivity

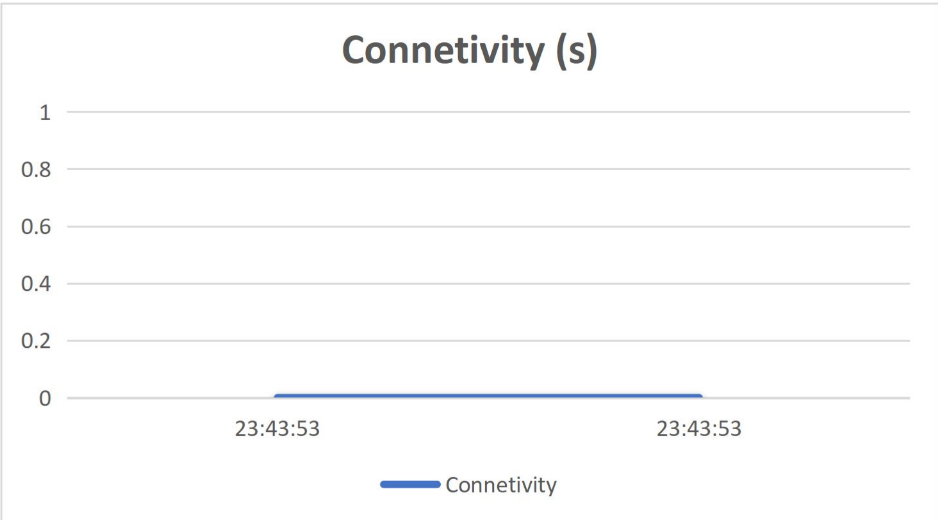
Adjust aircond value on IoT MQTT Panel

Device	IoT MQTT Panel	Node-RED Dashboard
Time	23:38:15	23:38:16



Adjust aircond value on Node-RED Control Tab

Device	IoT MQTT Panel	Node-RED Dashboard
Time	23:43:53	23:43:53



Conclusion

The IoT office monitoring project showcases the potential of modern technology in creating a smart, responsive, and efficient office environment. By using tools like Node-RED for data management, InfluxDB for storage, MQTT for communication, and Telegram for remote access, the system offers a comprehensive solution for monitoring and managing office spaces.

This project effectively simulates real-time data collection and processing for key environmental factors such as temperature, humidity, pressure, and smoke levels. Automated controls adjust equipment like air conditioning, lighting, and alarms based on sensor readings, improving both the efficiency and safety of the workspace. The integration with Telegram enables easy remote management, allowing users to control systems and receive updates from anywhere.

A major focus has been on scalability, ensuring the system can handle increasing data loads without losing performance. The reliability of data transmission and the system's low latency make real-time monitoring accurate and timely, which is essential for maintaining optimal office conditions.

Overall, this project demonstrates how IoT technologies can be seamlessly integrated to create a smart office monitoring system that is both functional and adaptable to various needs. It is a practical, easy-to-implement solution that sets a solid foundation for future enhancements and wider applications in IoT-driven office management.